

CSCI 5403

Introduction to Computing Security

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (68.9 / 100)	55.8 / 100

56	93	+47
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces core concepts in cybersecurity including confidentiality, integrity, authentication, risk management, and adversarial thinking. The concepts will be applied to both traditional information technology (IT) systems and cyber physical systems (CPS). At the conclusion of the course, students should have a solid foundation in cybersecurity and hands-on experience. Students must have access to either native or virtual machines running on Windows OR Linux (Parrot, Kali, Ubuntu).

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	68.9	100.0	+31.1	HIGH
Applied Scientist	61.6	100.0	+38.4	HIGH
Machine Learning Engineer	59.6	96.4	+36.9	HIGH
MLOps Engineer	52.7	95.0	+42.3	HIGH
ML Engineer	48.8	77.5	+28.6	MEDIUM
Data Scientist	43.3	90.2	+46.9	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

68.9 → 100.0 (+31.1 pts)

following data robotics engineering models learning software technical

Applied Scientist

61.6 → 100.0 (+38.4 pts)

computational literature problems models technical data learning engineering

Machine Learning Engineer

59.6 → 96.4 (+36.9 pts)

data web engineering sets learning tools protection business

MLOps Engineer

52.7 → 95.0 (+42.3 pts)

detection data model scientists learning machine media image

ML Engineer

48.8 → 77.5 (+28.6 pts)

media data learning machine users models model engineering

Data Scientist

43.3 → 90.2 (+46.9 pts)

data sets scripting different social engineering people problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.